

Multiple Sequence Alignment of Five Protein Sequences

CodeAlpha Bioinformatics Internship – Task 2

For this analysis, five protein sequences were selected and compared using Multiple Sequence Alignment (MSA). The sequences were aligned using Clustal Omega to identify conserved regions, similar residues, variable regions and sequence differences among the selected proteins.

Multiple Sequence Alignment is a bioinformatics technique used to compare three or more biological sequences simultaneously. It helps identify conserved amino-acid positions and regions, substitutions, insertions and deletions represented by gaps.

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CLUSTAL O(1.2.4) multiple sequence alignment

sequence3      LPISMDEGDSFLVKDSLGENK---IPKNPSKVILDGLDITFDALKLNDKVVGVPAK      56
sequence1      -----
sequence2      ---MDFNKEELEKMKSVIENAKKKEVAGAKPHI-TAAEGK-----LHNMIVDL---D      45

sequence3      NLPKYLQQFKNKPSVGGVQQVDFEAINALKPDLIISGRQSKFYDKLKEIAPTFLVGLDN      116
sequence1      -----
sequence2      NVVKK-----VQAAQSEAK-VVSQYHELVVQARDDFKRELDSTITPEVLPGWKG      92

sequence3      ANFLSSFENNVLVSAKLYGLEKEALEKISDIKNEIEKAKSIVDEDKKALIILTNSN---K      173
sequence1      -----
sequence2      MSVSDLA--DKLSTDDLNSLIAHAHRRIDQLNRELAEQKATE---KQHITLAEKQKLEE      147

sequence3      ISAFGPQS-----RFGIHDVLGIN      193
sequence1      -----
sequence2      KRAFDSAVAKALEHHRSEIQAEQDRKIEEVRDAMENEMRTLRRQAAAHDLRDVLRVQ      207

sequence3      AVD-----ENIKVGTGKKSINSEFILEKNPDYIFVVDRN-----      227
sequence1      -----MKKLLVFLLLISLFAV-----      17
sequence2      EQELKSEFEQNLSEKLSEQLQFRRLSQEQVDNFTLDINTAYARLRGIEQAVQSHAVAE      267
                      .   * *   .   :

sequence3      -----VILGN-----KERAQGILDNALVAKTKAA      251
sequence1      -DWSGKINWA---MNYNIAQSIK-----SSDKLIMVDIALTTCPCK      56
sequence2      EARKAHQLWLVSVEALKYSMTSSAETPTIPLGSAVEAIAKNCSDNE--FTQALTAAPPE      325
                      .       .   * * . :

sequence3      QNKKIYLDPEYWYLASGN-----GLESKTMILEIKNAVK-----      287
sequence1      YLAEKVYTDPESKYINDNFVAVFYLDQ----DSLPTVQNYFTGSTPTILFLKPN---      109
sequence2      SLTRGVYSEETLR---ARFYAVQKLARRVAMIDETRNSLYQYFLSYLQSLLLFPPQQLK      381
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Figure 1. Clustal Omega multiple sequence alignment of the selected protein sequences.

Alignment Result

The Clustal Omega output shows the five sequences arranged according to their sequence similarity. Gaps have been introduced at several positions to optimize the alignment. The alignment contains both conserved and variable regions, indicating that the sequences share some similarities while also showing sequence divergence.

The conservation symbols shown below the alignment provide additional information. An asterisk (*) represents an identical residue at that position, a colon (:) represents strongly similar residues, and a period (.) represents weakly similar residues. These symbols are useful for identifying conserved positions.

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sequence1  ----- 222
sequence4  ----- -LKYGVHEIIYTMLASLMNKDGVLISEGQGFMTREFLKSLRKPFQDFM 162
sequence5  FKFLKNNLDAAKDSWKTLRITDAVTLKEGKEL----- 175
sequence2  HAVAEERARKAHQLWLSVEALKYSMTSSAETP-----TIPLGS-- 300
sequence3  -----DWSGKINWA-----MNYNIAQSIK----- 37

sequence1  -----VVDNRNVLGNKERAQGILDNALVAKTKAAQNKKIYLDPEY----- 263
sequence4  EPKFEEFAVKFNALEDDSDLA-----IFIIVII 190
sequence5  -----KVETVTPQK----- 184
sequence2  -----AVEAIKANCSNE--FTQALTAIPPELTRGVYSEETLR---ARFYAVQK 346
sequence3  -----SSDKLIMVDIALTTCPPCKYLAEKVYTDPEVSKYINDNFVAVFY 81

sequence1  -----WYLASGNGLKESLKTML-----EIKNAVK----- 287
sequence4  LS-----GDRPGLLVNPKPI-----EDIQDNLQALELQ-LKLNHPE- 225
sequence5  ----- 184
sequence2  LARRVAMIDETRNSLYQYFLSYLQSLLLFPQQKPPPELCPEDINTFKLLSYASYCIEH 406
sequence3  LADQ---DSLPTVQNYFTGSTPTILFLKPN-----GELVYSMIGAR-----PPQ 123

sequence1  ----- 287
sequence4  -SSQLFAKLLQKMTDL-RQI-----VTEHVQLLVIKKTETDMS--LHPLLQEIY 271
sequence5  ----- 184
sequence2  GDLELAAKFVNQLKGESRRVAQDWLKEARMTLETKQIVEILTAYASAVGIGTTQVQPE-- 464
sequence3  AFLQIITKNVNNKYKGNK----- 140

sequence1  ---- 287
sequence4  KDLY 275
sequence5  ---- 184
sequence2  ---- 464
sequence3  ---- 140

```

Figure 2. Detailed Clustal Omega alignment showing conserved, similar and variable regions.

The detailed alignment demonstrates conserved amino-acid stretches along with regions containing substitutions and gaps. The displayed sequences also differ in length, with sequence2 reaching approximately 464 residues, sequence1 287 residues, sequence4 275 residues, sequence5 184 residues, and sequence3 140 residues. The gaps help maintain positional correspondence between sequences of different lengths.

Overall, the presence of conserved residues suggests that some regions may be important for maintaining protein structure or function, while variable regions may reflect differences between the selected sequences.

Conclusion

Multiple Sequence Alignment of five protein sequences was successfully performed using Clustal Omega. The alignment revealed conserved and variable amino-acid regions, along with gaps representing insertions or deletions. The conservation symbols helped identify identical and similar residues. Overall, MSA provides a useful approach for comparing protein sequences and identifying regions that may have structural or functional significance.